CS24305 — THEORY (3.0 CR) MODULE III NOTES BIT MESRA | NEP SCHEME

Module 3: Error Control & Data Link Protocols

CRC-32, Hamming Code, Stop-and-Wait, GBN, Selective Repeat, HDLC

Module III: Error Control & Data Link Protocols — Topics Covered

1. Cyclic Redundancy Check (CRC-16/32 Modulo-2)
2. Hamming Distance & (7, 4) Error Correction
3. Sliding Window ARQ: Stop-and-Wait, GBN, SR
4. HDLC Protocol & Bit Stuffing Mechanics
5. Frequency and Time Division Multiplexing (FDM/TDM)

1. Cyclic Redundancy Check (CRC)

Given data word $D(x)$ of k bits and generator polynomial $G(x)$ of degree n:

1. Append n zero bits to $D(x)$ to form dividend $D(x) \cdot 2^n$.
2. Perform binary Modulo-2 division (XOR) by generator polynomial $G(x)$.
3. The n-bit remainder $R(x)$ is the Frame Check Sequence (FCS). Transmit $T(x) = D(x) \cdot 2^n \oplus R(x)$.

2. Sliding Window ARQ Protocols Comparison

Protocol	Sender Window (W_s)	Receiver Window (W_r)	Efficiency (η)	Retransmission Behavior
Stop-and-Wait	1	1	$\frac{1}{1+2a}$ where $a = \frac{T_{\text{prop}}}{T_{\text{trans}}}$	Retransmits single frame upon timer expiry.
Go-Back-N (GBN)	$2^k - 1$	1	$\frac{W_s}{1+2a}$	Discards out-of-order frames; retransmits entire window from lost frame.
Selective Repeat	2^{k-1}	2^{k-1}	$\frac{W_s}{1+2a}$	Buffers out-of-order frames; retransmits **only the damaged frame** (NAK).